

```
In [1]: import numpy as np
import pandas as pd
import seaborn as sns
```

```
In [15]: df = sns.load_dataset('titanic')
```

```
In [4]: df.isnull().sum()
```

```
Out[4]: survived      0
pclass      0
sex          0
age        177
sibsp       0
parch       0
fare        0
embarked     2
class        0
who          0
adult_male   0
deck       688
embark_town   2
alive        0
alone        0
dtype: int64
```

```
In [5]: df.shape
```

```
Out[5]: (891, 15)
```

```
In [6]: df.describe()
```

```
Out[6]:
```

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

```
In [7]: df.dtypes
```

```
Out[7]: survived      int64
        pclass        int64
        sex           object
        age           float64
        sibsp         int64
        parch         int64
        fare          float64
        embarked      object
        class         category
        who           object
        adult_male     bool
        deck          category
        embark_town    object
        alive         object
        alone         bool
        dtype: object
```

```
In [16]: df['age']=df['age'].fillna(df['age'].mean())
```

```
In [10]: df.isnull().sum()
```

```
Out[10]: survived      0
        pclass        0
        sex           0
        age           0
        sibsp         0
        parch         0
        fare          0
        embarked      2
        class         0
        who           0
        adult_male     0
        deck          688
        embark_town    2
        alive         0
        alone         0
        dtype: int64
```

```
In [17]: df['embarked']=df['embarked'].fillna(df['embarked'].mode()[0])
```

```
In [18]: df=df.drop('deck',axis=1)
```

```
In [20]: df['embark_town']=df['embark_town'].fillna(df['embark_town'].mode()[0])
```

```
In [21]: df.isnull().sum()
```

```
Out[21]: survived      0
         pclass        0
         sex           0
         age           0
         sibsp         0
         parch         0
         fare          0
         embarked      0
         class         0
         who           0
         adult_male    0
         embark_town   0
         alive         0
         alone         0
         dtype: int64
```

```
In [23]: df['age']=df['age'].astype(int)
```

```
In [27]: df['fare']=(df['fare']-df['fare'].min()/(df['fare'].max()-df['fare'].min()))
         # data normalization : Scaling data to fixed range., need: Prevents large values do
```

```
In [28]: df.head(10)
```

```
Out[28]:
```

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult
0	0	3	male	22	1	0	0.014151	S	Third	man	
1	1	1	female	38	1	0	0.139136	C	First	woman	
2	1	3	female	26	0	0	0.015469	S	Third	woman	
3	1	1	female	35	1	0	0.103644	S	First	woman	
4	0	3	male	35	0	0	0.015713	S	Third	man	
5	0	3	male	29	0	0	0.016510	Q	Third	man	
6	0	1	male	54	0	0	0.101229	S	First	man	
7	0	3	male	2	3	1	0.041136	S	Third	child	
8	1	3	female	27	0	2	0.021731	S	Third	woman	
9	1	2	female	14	1	0	0.058694	C	Second	child	

◀  ▶

```
In [33]: df['sex'] = df['sex'].map({'male':0 , 'female':1})
         # convert categorical data into numerical data : ML algorithms work on numeric data
```

```
In [30]: df.head()
```

Out[30]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adult_male
0	0	3	0	22	1	0	0.014151	S	Third	man	True
1	1	1	1	38	1	0	0.139136	C	First	woman	False
2	1	3	1	26	0	0	0.015469	S	Third	woman	False
3	1	1	1	35	1	0	0.103644	S	First	woman	False
4	0	3	0	35	0	0	0.015713	S	Third	man	True

Theory for DSBDA Practical — Data Wrangling I

1. What is Data Wrangling?

Data Wrangling means:

Cleaning data Transforming data Organizing raw data into useful format

It is also called:

Data Cleaning Data Preprocessing Why Data Wrangling is Needed?

Real-world datasets contain:

Missing values Incorrect data types Duplicate records Noisy data Inconsistent formatting

Data wrangling improves:

Data quality Accuracy Machine learning performance

2. What is Data Preprocessing?

Data preprocessing is the process of preparing raw data before analysis or machine learning.

Main steps:

Handling missing values Removing duplicates Type conversion Encoding categorical variables Normalization/Scaling

3. What is Pandas?

pandas is a Python library used for:

Data manipulation Data analysis Handling tabular data

Main Data Structures:

Series → One-dimensional DataFrame → Two-dimensional table

4. What is NumPy?

NumPy is a Python library used for:

Numerical operations Arrays Mathematical calculations

Advantages:

Faster computations Memory efficient

5. What is a Dataset?

Dataset is a collection of related data.

Example: Titanic dataset contains:
Passenger names Age Gender Fare Survival status
Types of Datasets Type Description
Structured Tables, rows, columns Unstructured Images, videos Semi-structured JSON, XML

Titanic dataset is structured data.

6. What is CSV File?

CSV = Comma Separated Values

Features:

Stores tabular data Lightweight Easy to read

Example:

Name, Age, Gender John, 22, Male

7. What is a DataFrame?

DataFrame:

Two-dimensional tabular structure Contains rows and columns Similar to Excel sheet

Features:

Easy filtering Data analysis Statistical operations

8. Understanding Dataset Dimensions Shape of Dataset

`df.shape`

Returns:

(rows, columns)

Example:

(891, 12)

Meaning:

891 records 12 features

9. Types of Variables

Variables are columns/features in dataset.

A. Numerical Variables

Contain numbers.

i) Integer

Whole numbers.

Example:

PassengerId Survived ii) Float

Decimal values.

Example:

Age Fare B. Categorical Variables

Contain categories/text.

Example:

Male/Female City names

Datatype:

object

10. Missing Values

Missing value means data is absent.

Represented as:

NaN Null

Example:

Age = NaN Causes of Missing Values Data entry errors Incomplete surveys System failures Problems Due to Missing Values Incorrect analysis Reduced accuracy Errors in machine learning Methods to Handle Missing Values A. Remove Missing Data Row deletion `dropna()`

Used when missing values are few.

B. Replace Missing Values Mean Imputation

Used for numerical data.

Formula:

$$\text{Mean} = \sum x/n$$

Median Imputation

Middle value of sorted data.

Good for skewed data.

Mode Imputation

Most frequent value.

Used for categorical data.

11. describe() Function Theory

Used to generate statistical summary.

Provides:

Count Mean Standard deviation Minimum Maximum Quartiles Mean

Average value.

Formula:

$$x = \sum x/n$$

Standard Deviation

Measures spread of data.

High SD:

Data widely spread

Low SD:

Data close to mean Quartiles

Divide data into 4 parts:

Q1 = 25% Q2 = Median Q3 = 75%

12. Data Types in Python/Pandas Type Meaning int64 Integer float64 Decimal object
String/Text bool True/False

13. Type Conversion

Changing one datatype to another.

Example:

astype(int) Why Type Conversion Needed? Data consistency Proper calculations Machine learning compatibility 14. Categorical Variables

Variables containing labels/categories.

Example:

Male/Female Yes/No

Machine learning models cannot directly process text categories.

So they must be converted to numbers.

15. Label Encoding

Converts categories into numbers.

Example:

Male = 0 Female = 1

Advantages:

Simple Memory efficient

Disadvantage:

Creates false ordering 16. One Hot Encoding

Creates separate binary columns.

Example:

Gender_Male Gender_Female 1 0 0 1

Advantages:

No false ordering

Disadvantages:

More columns created 17. Data Normalization

Normalization scales data into fixed range.

Usually:

0 to 1 Why Normalization Needed?

Features may have different ranges.

Example:

Age → 1 to 80 Fare → 1 to 500

Large values dominate calculations.

Normalization balances them.

Min-Max Normalization

Formula:

$$x' = (x - x_{\min}) / (x_{\max} - x_{\min})$$

Where:

x = original value x_{min} = minimum value x_{max} = maximum value
18. Difference Between Normalization and Standardization
Normalization Scales between 0 and 1
Standardization Mean becomes 0
Uses min-max Uses standard deviation
Sensitive to outliers Less sensitive

19. Duplicate Data

Repeated rows in dataset.

Problems:

Increases dataset size Biased analysis

Detection:

`duplicated()`

Removal:

`drop_duplicates()`

20. Importance of Data Cleaning

Clean data helps in:

Better visualization Accurate predictions Faster processing Improved ML models
21. Machine Learning and Preprocessing

Machine learning algorithms require:

Numeric data Clean data No missing values Proper scaling

Therefore preprocessing is essential.

22. Difference Between Structured and Unstructured Data

Structured	Unstructured
Table format	No fixed format
Easy analysis	Difficult analysis
CSV, Excel	Images, videos

23. Important Viva Questions Basic What is Data Wrangling?

Process of cleaning and transforming data.

What is preprocessing?

Preparing raw data for analysis.

Why missing values handled?

To improve accuracy.

Why normalization used?

To scale features equally.

Intermediate Difference between Label Encoding and One Hot Encoding? Label Encoding One Hot Encoding Single column Multiple columns Numeric labels Binary vectors Why categorical variables converted?

ML models require numerical input.

Difference between DataFrame and Series? Series → 1D DataFrame → 2D 24. Frequently Asked External Questions What happens if missing values are not handled?

Analysis becomes incorrect.

Why use Pandas?

Easy and powerful data analysis.

Which function checks null values? `isnull()` Which function shows dataset statistics? `describe()` Which function changes datatype? `astype()` Difference between loc and iloc? loc → label based iloc → index based 25. Real Life Applications of Data Wrangling

Used in:

Banking Healthcare E-commerce Social media analytics Fraud detection